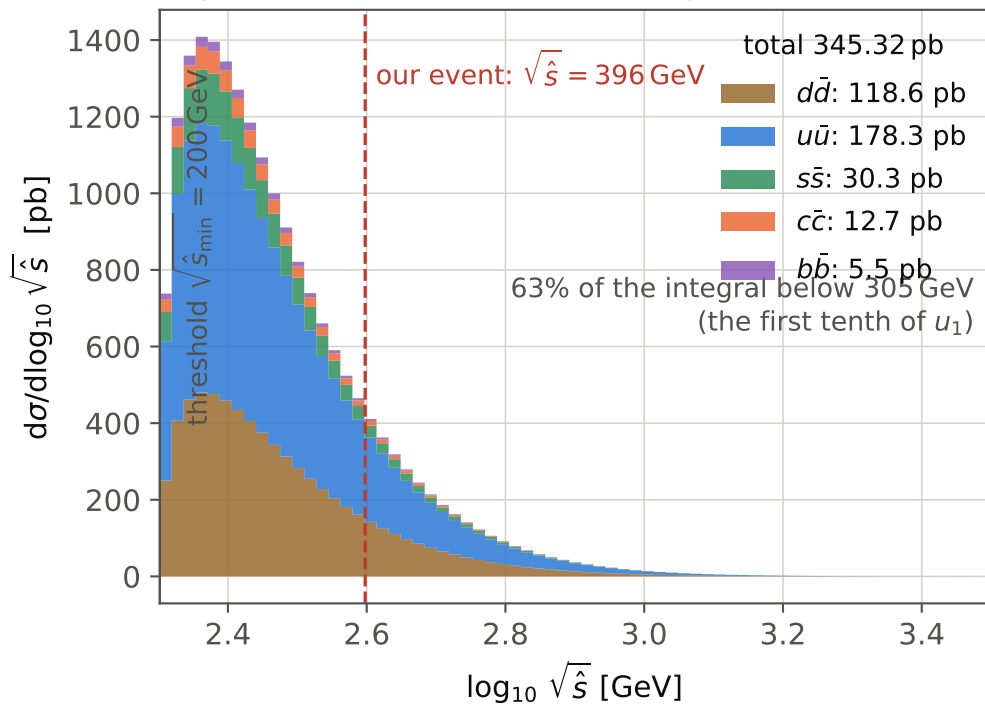
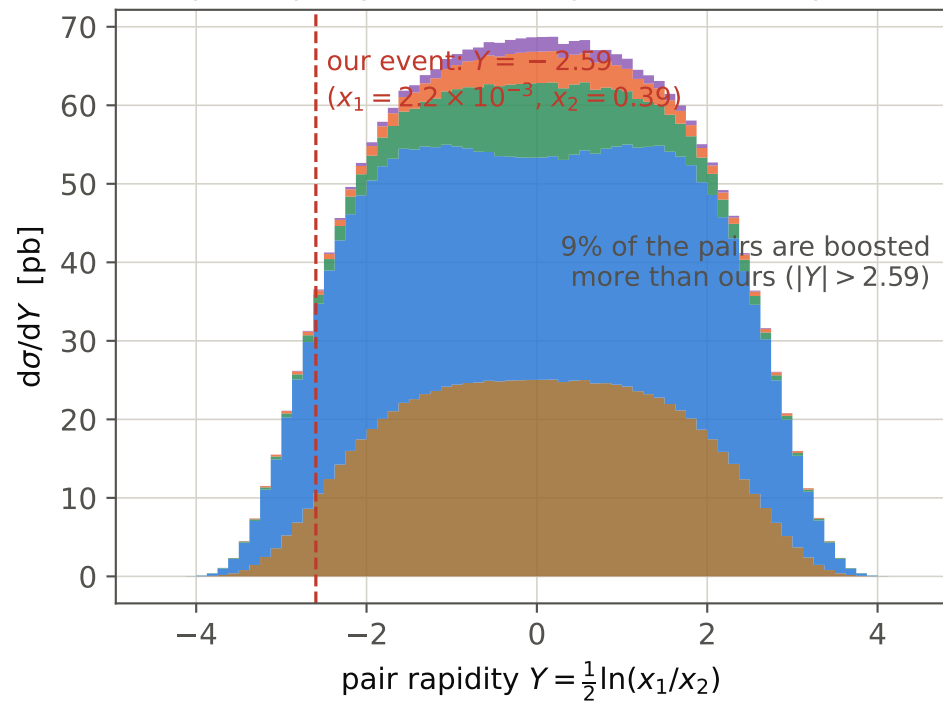


Where the 345 pb come from: the integrand of the factorisation formula, one Sobol pass of 2^{18} points

(a) the pair mass: most of the rate sits just above the threshold

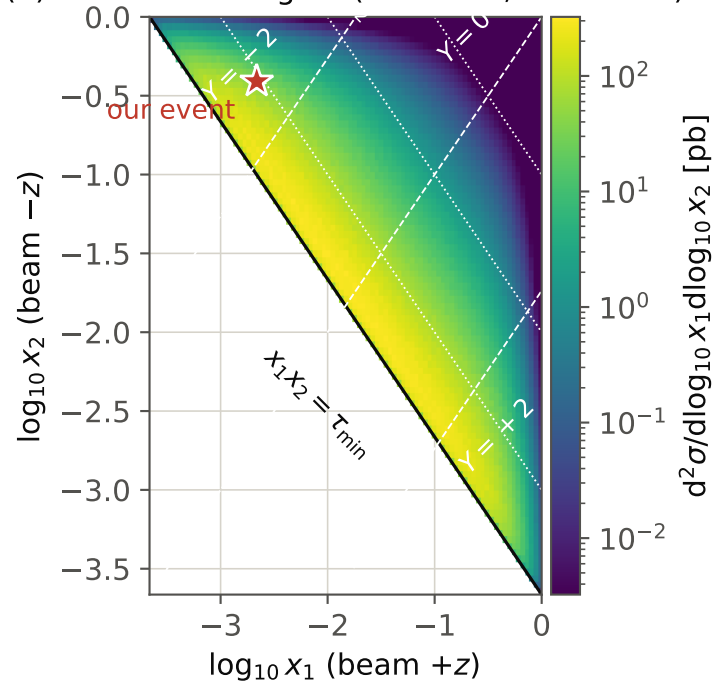


(b) the pair rapidity: where along the beam the pair is born

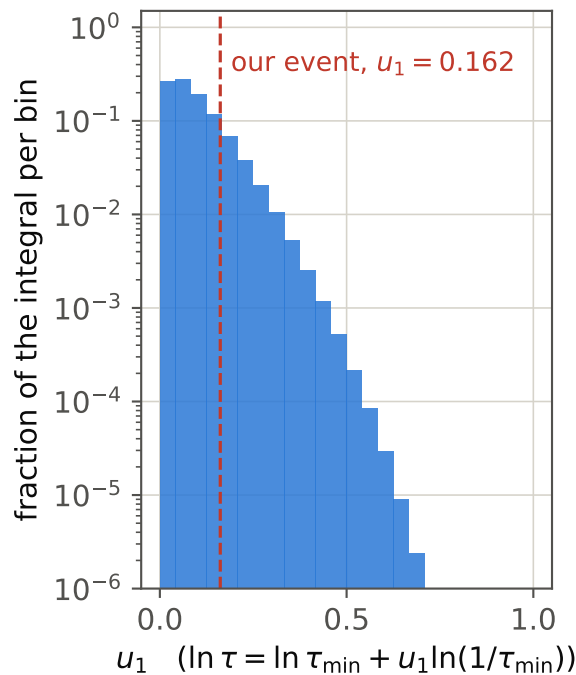


Better variables: the triangle in $(\log x_1, \log x_2)$ and why $\ln \tau$ is sampled

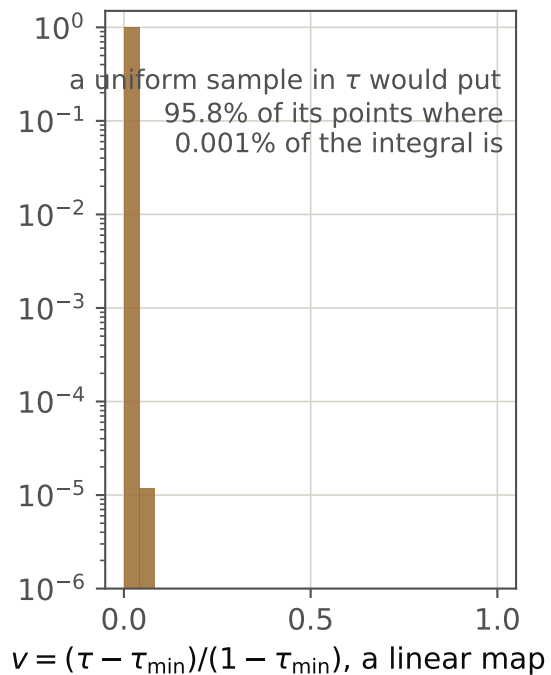
(a) the allowed region (dotted: τ , dashed: Y)



(b) linear in $\ln \tau$: first bin 26%

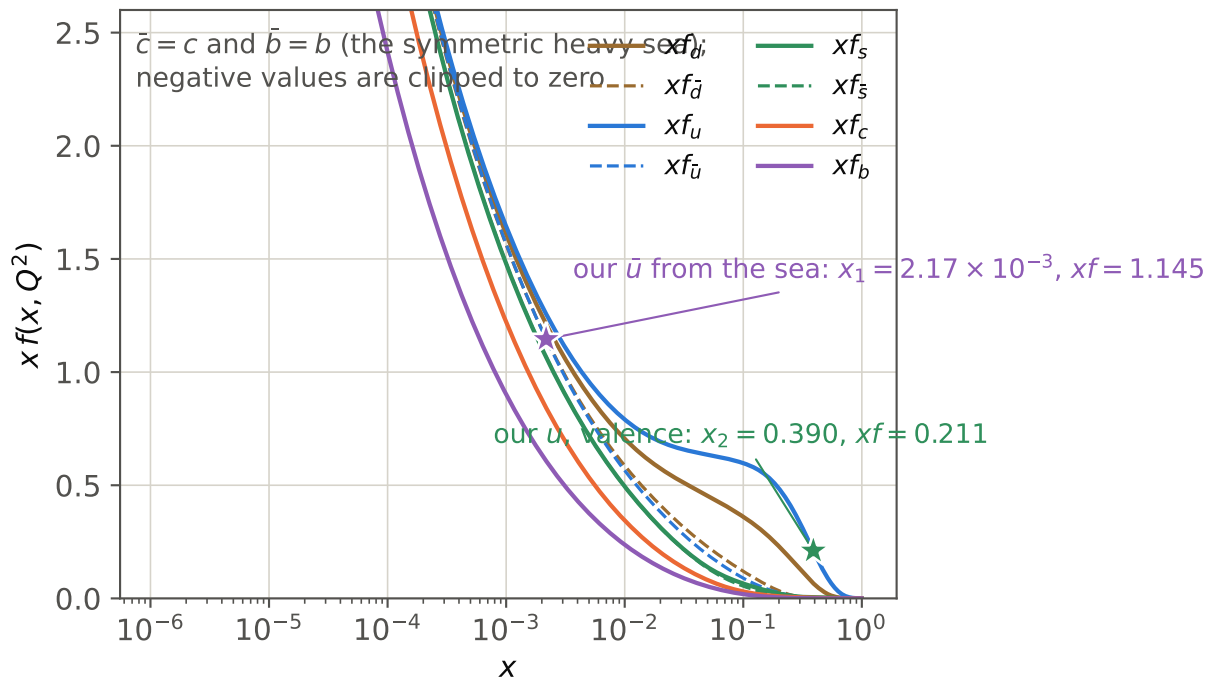


(c) linear in τ : first bin 100.00%

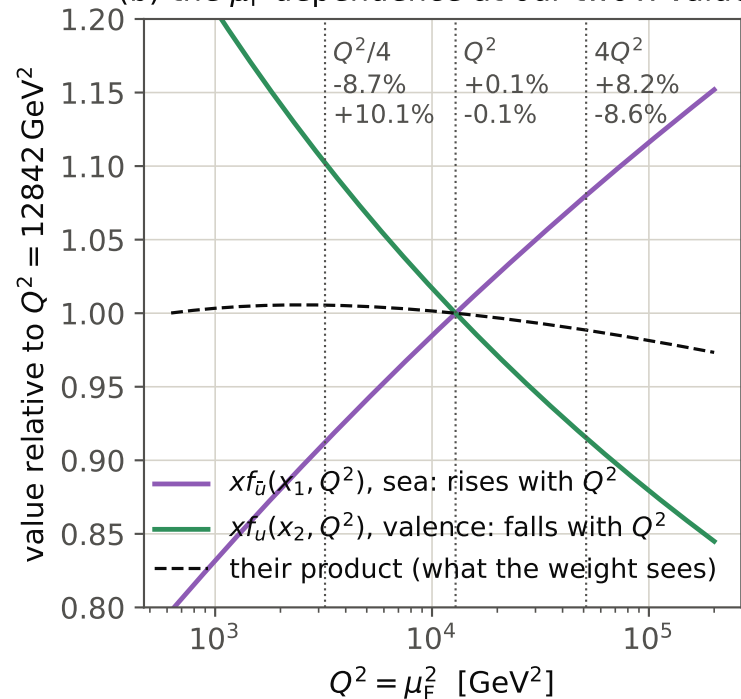


The PDFs of the embedded table (NNPDF3.1 NNLO, member 0) at the event's scale $Q^2 = 12842 \text{ GeV}^2$

(a) $xf(x, Q^2)$: sea rising at small x , valence bump at $x \sim 0.2$

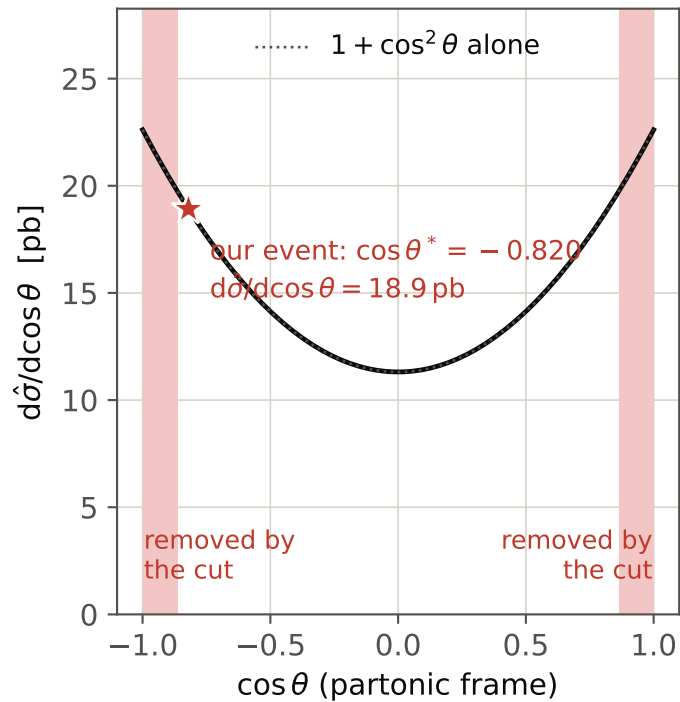


(b) the μ_F dependence at our two x values

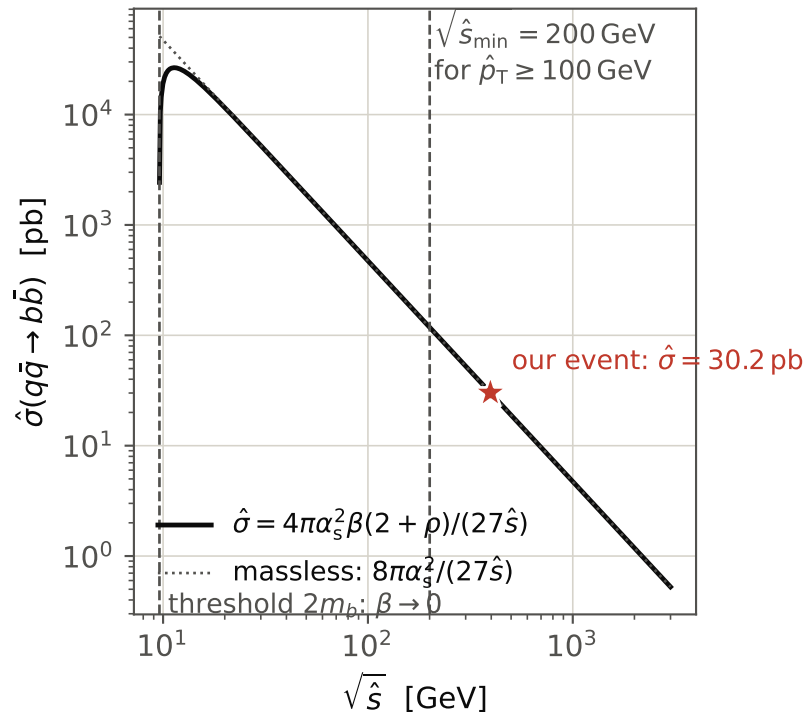


The Born matrix element $q\bar{q} \rightarrow g^* \rightarrow b\bar{b}$ and what the $\hat{p}_T \geq 100$ GeV cut does to it

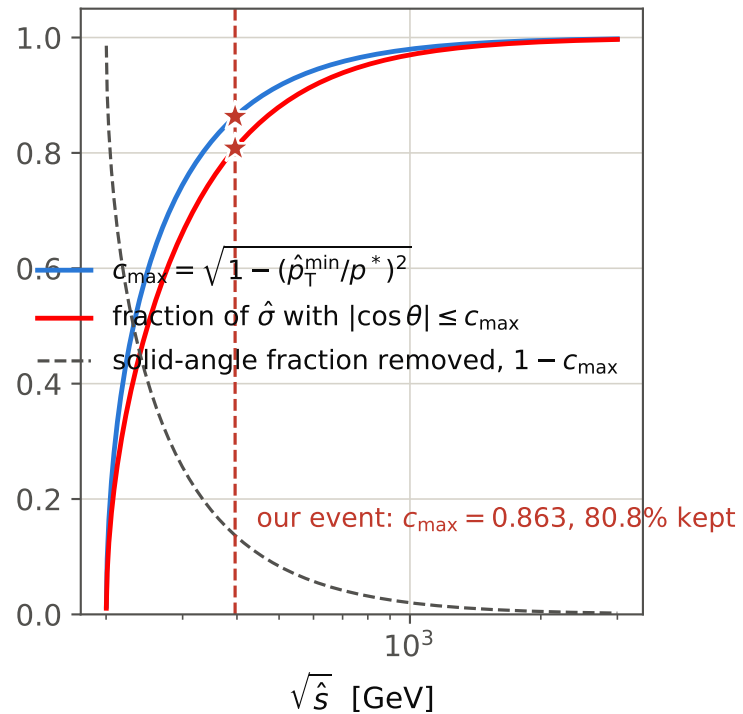
(a) at $\sqrt{\hat{s}} = 396$ GeV: $|\cos \theta| \leq 0.863$ keeps 80.8%



(b) the partonic cross section at fixed $\alpha_s = 0.1143$

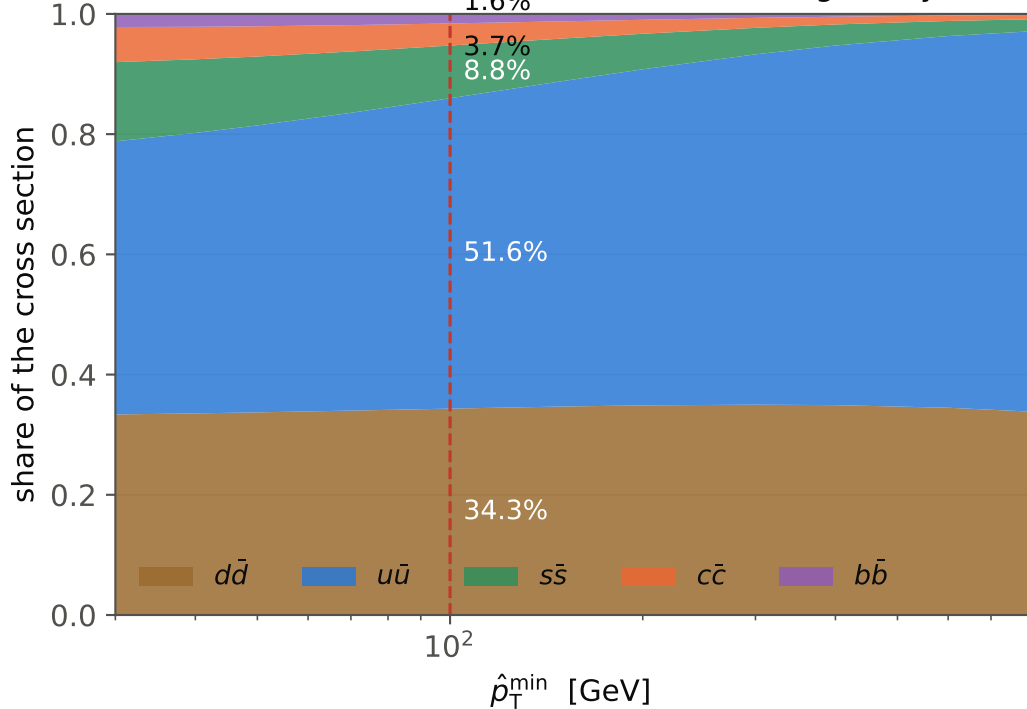


(c) the cut on $\hat{p}_T$ is a cut on the angle

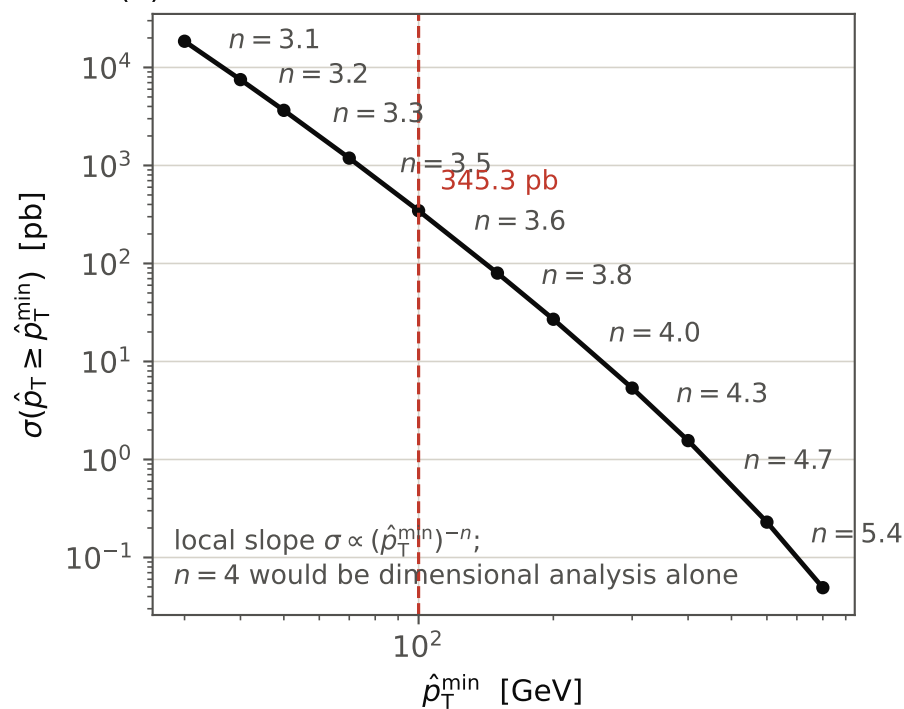


Which quarks make the pair: the share of each incoming flavour (single Sobol passes of the standalone script)

(a) the valence u wins as the cut rises; the incoming b stays below 2%

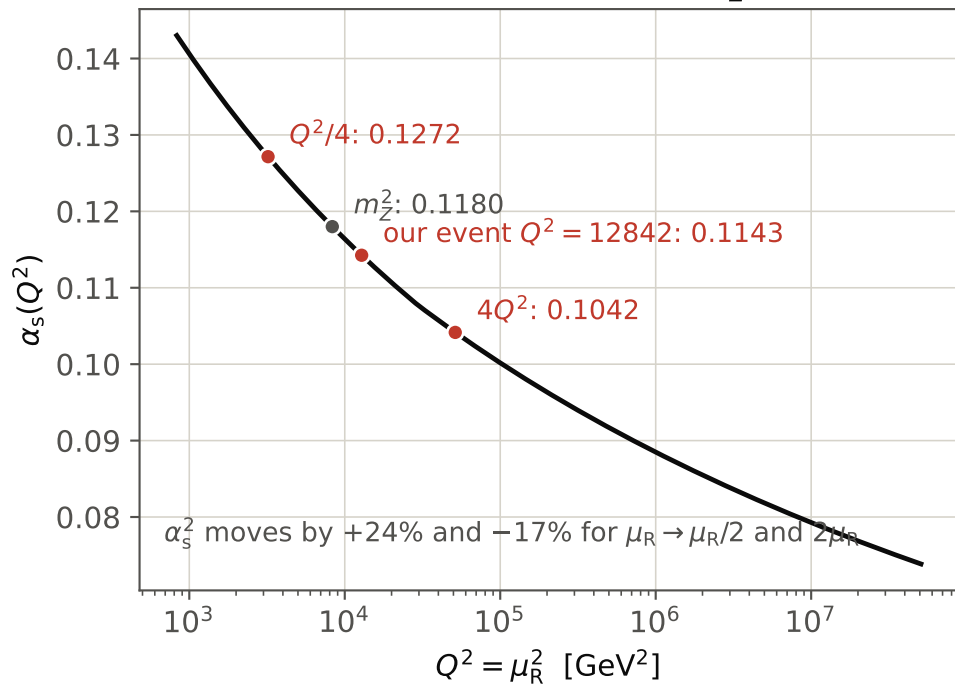


(b) the total: a factor 54 between 30 and 100 GeV

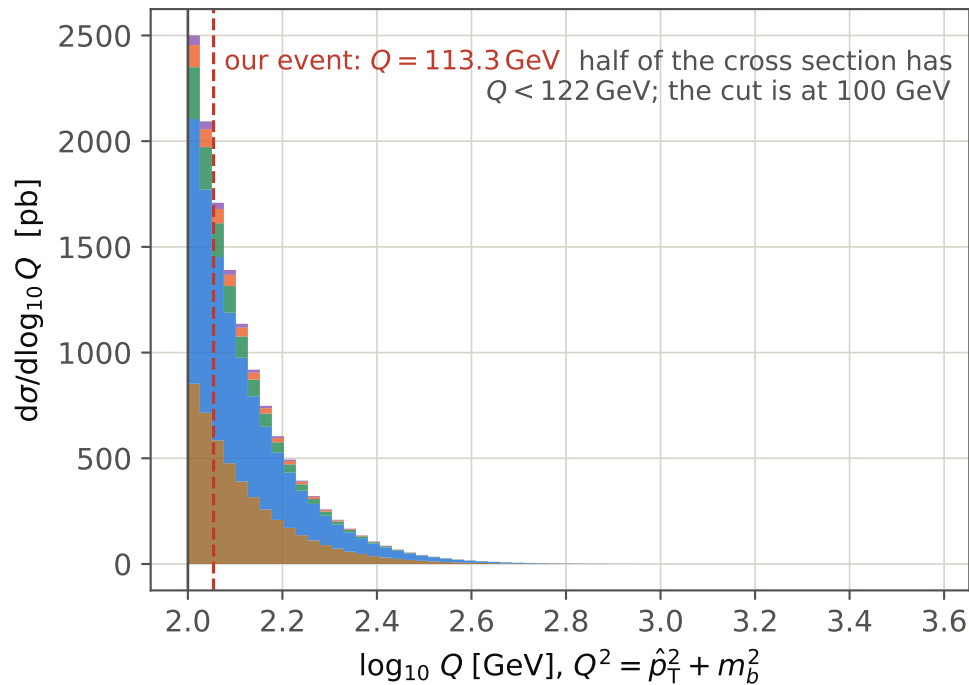


The scale of the hard process: $\mu_R^2 = \mu_F^2 = \hat{p}_T^2 + m_b^2$ and PYTHIA's second-order α_s

(a) the embedded table (TableAlphaS): $\alpha_s(m_Z^2) = 0.118$, two loops

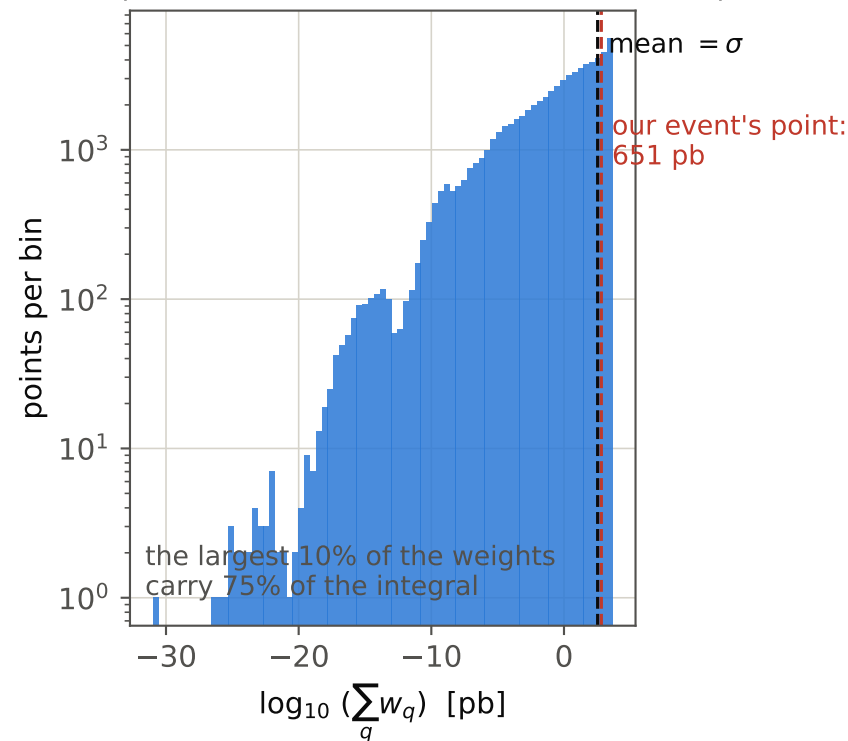


(b) the scales the integral actually visits

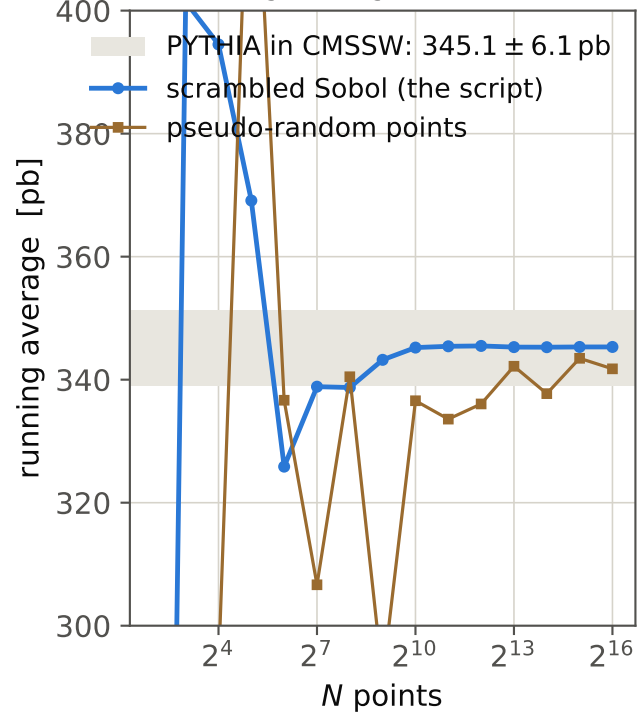


From formula to number: the weights of one Sobol pass, the running average and the error against N

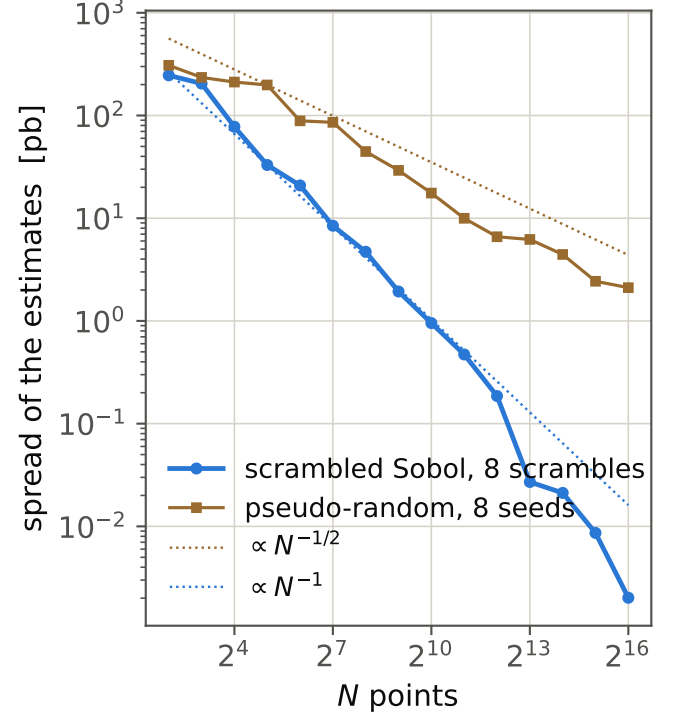
(a) one pass: median 1.0, mean 345.3, max 4576 pb



(b) the running average settles at 345.32 pb

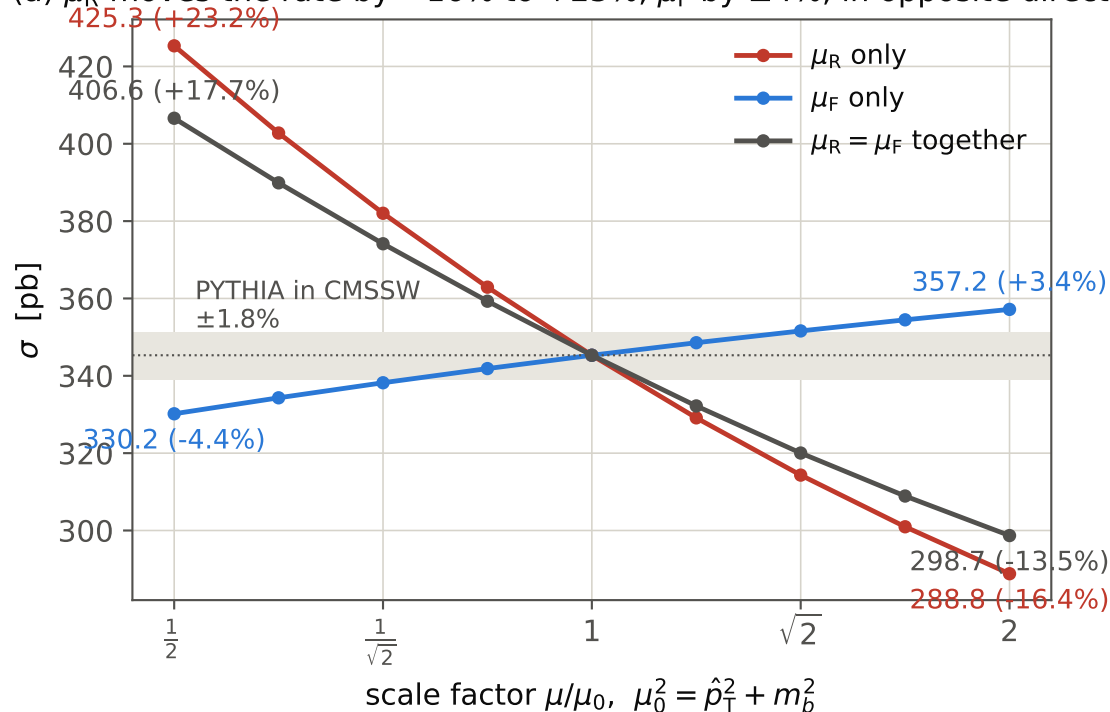


(c) spread at 2^{16} : 0.002 (Sobol), 2.11 pb (rand)



Scale variations of the leading-order cross section (qqbar_bbbar_xsec.py through scan_scales.py)

(a) μ_R moves the rate by -16% to $+23\%$, μ_F by $\pm 4\%$, in opposite directions



(b) the seven-point envelope: 289 to 425 pb, $+23\%$ to -16%

